From Singularities to Asymptotic Bounds

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Abstract

Combinatorial sequences such as partition numbers (see Example 5.1) have been studied for a long period of time. Yet often times, it is exceptionally difficult to determine a closed formula for such sequences. Though there has been recent success in deriving a closed form for the partition numbers using algebraic methods (see [1]), it is computationally more efficient to determine accurate asymptotic bounds on such sequences. Notably, Ramanujan and Hardy provided an asymptotic estimate of partition numbers $p(n)$ given by $p(n) \sim \frac{1}{4n\sqrt{3}} \exp\left(\pi\sqrt{\frac{2n}{3}}\right)$ (see [4]). This report will present three different techniques on how such bounds can be found based on singularities of generating functions. Although these methods have been explored in the standard work "Analytic Combinatorics" by Philippe Flajolet and Robert Sedgewick (see [2]), this report has generalized some results like Pringsheim's Theorem (Theorem 3.2) and formalized the proofs of these, which were only given as rough sketches in [2]. Additionally, these methods will be applied to Fibonacci sequences as well as another recursively defined sequence, to examine the accuracy of the approximations.

1 Introduction

Generating functions play a central role in combinatorics and characterize a sequence in a special way. By nature, generating functions are nothing more than a power series and by extending generating functions into the complex plane, a lot of tools from complex analysis may be applied to tackle problems and have proven to be useful. For example, Cauchy's integral formula allows for an alternative characterization of the coefficients of generating functions and may even provide a closed formula, if the integral can be calculated. Though these methods have been used historically by the likes of Ramanujan to establish the asymptotic estimates on the partition numbers (see [4]), it has only been in 2009 that Philippe Flajolet and Robert Sedgewick published the book "Analytic Combinatorics" ([2]), which summarizes various results from complex analysis that are applicable to combinatorics. They are now considered one of the main pioneers of this field [6].

2 Preliminaries

We will first introduce a few fundamental combinatorial definitions.

Definition 2.1 (Combinatoral class)**.** *A combinatorial class is a pair $(\mathcal{A}, |\cdot|)$, where $\mathcal{A}$ is a countable set of combinatorial objects, $|\cdot| : \mathcal{A} \to \mathbb{N}$ is a function called the size function and $\mathcal{A}_n = \{x \in \mathcal{A} \colon |x| = n\}$ is finite for all $n \in \mathbb{N}$.*

Definition 2.2 (Generating function)**.** *Let a_n be the number of objects of size n in $\mathcal{A}$. Then the power series*

$$G(z) = \sum_{n=0}^{\infty} a_n z^n = \sum_{\alpha \in \mathcal{A}} z^{|\alpha|}$$

is the (ordinary) generating function of the class $\mathcal{A}$.

We highlight these definitions with some examples.

Example 2.3 (Fibonacci numbers)**.** One of the most basic combinatorical sequences is given by the Fibonacci numbers, which is defined by $f_0 := 0$, $f_1 := 1$ and the recurrence relation $f_n := f_{n-1} + f_{n-2}$ for $n \geqslant 2$. A simple calculation using the recurrence relation (see [7]) shows that the generating function of the Fibonacci numbers is given by

$$G(z) = \sum_{n=0}^{\infty} f_n z^n = \frac{z}{1 - z - z^2}.$$

Furthermore, the Fibonacci numbers also admit a closed form known as the Binet-Euler formula

$$f_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right].$$

Example 2.4 (Generating function using recursion formula)**.** Many sequences are defined by some recursion formula. Another example is given by the recursive relation $a_0 = a_1 = 1$ and $a_n = 4a_{n-1} - 4a_{n-2} + \binom{n}{2} 2^n + 1$. The generating function is then given by

$$G(z) = \frac{1 - 10z + 44z^2 - 84z^3 + 80z^4 - 32z^5}{(1 - z)(1 - 2z)^5},$$

whereby a derivation can be found here [3].

In the following, we will show how asymptotic estimates for sequences without explicit closed forms can be derived with tools from complex analysis.

3 Methods

We begin with an important relationship between singularities of holomorphic functions and its radius of convergence.

Proposition 3.1. *Let $f : \mathbf{C}\backslash S \to \mathbf{C}$ be holomorphic in $\mathbf{C}$ outside of a discrete set S of singularities. Let $z_0 \in \mathbf{C}\backslash S$. Then the radius of convergence of the Taylor series of f around z_0 is equal to the distance from z_0 to the nearest non-removable singularity in S, or $+\infty$ if no such non-removable singularity exists.*

Proof. First, assume that S contains at least one non-removable singularity. Let $z_0 \in \mathbb{C}\backslash S$ be arbitrary and denote the smallest distance from z_0 to a singularity point by $d_{z_0} := \min\{|z_0 - s| \colon s \in S, s \text{ non-removable singularity}\}$, whereby the minimum is attained for some $z^* \in S$, as S is discrete and contains isolated points. Furthermore, let R_{z_0} represent the radius of convergence of the Taylor expansion around z_0.

Clearly the relation $R_{z_0} \leqslant d_{z_0}$ holds, else the Taylor-expansion of f would could converge in z^* and coincide with the Laurent-expansion in z^*, so z^* cannot be a non-removable singularity, which is a contradiction. It remains to show that $R_{z_0} \geqslant d_{z_0}$ is true. So, let $\varepsilon > 0$ be arbitrary and consider $\tilde{d} := d_{z_0} - \varepsilon$, then there exists a bounded open set $U \subseteq \mathbb{C}$ with $z^* \notin U$ and $\overline{D(z_0, \tilde{d})} \subseteq U$ such that the restriction $f|_{U\backslash\tilde{S}} \colon U\backslash\tilde{S} \to \mathbb{C}$ is holomorphic, whereby $\tilde{S}$ only contains removable singularities or is the empty set. Note that $\tilde{S}$ is compact and of zero length, so by Painlevé's theorem (Exercise 3, Notes 4), there exists an extension of $\tilde{f} \colon U \to \mathbb{C}$ of $f|_{U\backslash\tilde{S}}$ to U that remains holomorphic. Since the Taylor-expansion of $\tilde{f}$ and $f|_{U\backslash\tilde{S}}$ agree on an open neighborhood around z_0, the Taylor-series are the same. So by Corollary 20 (Notes 3), the inequality $R_{z_0} > \tilde{d} = d_{z_0} - \varepsilon$ follows. As this is true for all $\varepsilon > 0$, this implies $R_{z_0} \geqslant d_{z_0}$ and one concludes the wanted equality $R_{z_0} = d_{z_0}$.

For the case that S does not contain any non-removable singularities and $d_{z_0} = \infty$, one can use the same argument as above to show that $R_{z_0} > K$ holds for any $K > 0$.

□

Naturally the question arises, whether the assumption of a discrete set S is necessary. From the proof, it can directly be seen that the assumption that S being compact and of zero-length suffices. Moreover, the case that $d_{z_0} = 0$ cannot occur, as f is holomorphic in z_0 by assumption.

In the context of combinatorics, one can refine the statement of Proposition 3.1, which is known as Pringsheim's theorem.

Theorem 3.2 (Pringsheim's Theorem)**.** *If a function $f \colon \mathbb{R} \to \mathbb{R}$ is analytic at the origin and has non-negative Taylor-coefficients and a radius of convergence R, then the point $z = R$ is a singularity of f.*

Proof. Consider the Taylor expansion around the origin and replace the real variable with a complex variable, i.e. $f(z) = \sum_{n=0}^{\infty} a_n z^n$. By Proposition 3.1 there exists a non-removable

singularity $z_0 \in \mathbb{C}$ with $|z_0| = R$. Given that the coefficients a_n are real and non-negative, the triangle inequality shows that $\left|\sum_{n=0}^{\infty} a_n z^n\right| \leqslant \sum_{n=0}^{\infty} a_n R^n$ holds true for $|z| = R$. Then, we have $|f(z_0)| \leqslant f(R)$ and by Theorem 12, Notes 4, we have $f(z) \to \infty$ as $z \to R$. This shows that the point $z = R$ is indeed a singularity of f. □

The proof given in (pg.256, [2]) is based on direct calculations, whereas the tools of complex analysis provide more intuition as to why the statement is true.

Once we know the singularities, we would like to get an asymptotic expansion for the sequence. From the definition of the radius of convergence (Notes 2, Definition 3), we have $\liminf_{n\to\infty} |a_n|^{-1/n} = R$, which is equivalent to $\limsup_{n\to\infty} |a_n|^{1/n} = \frac{1}{R}$. This describes the asymptotic behavior of the sequence a_n and is formalised in the following Corollary.

Corollary 3.3 (Exponential order)**.** *A sequence $(a_n)_{n\in\mathbb{N}}$ is of exponential order K^n*

$$a_n \bowtie K^n \quad \textit{iff} \quad \limsup |a_n|^{1/n} = K.$$

Furthermore this is both an upper bound and a lower bound, and for any $\epsilon > 0$:

(i) $|a_n| > (K - \varepsilon)^n$ infinitely many times often.

(ii) $|a_n| < (K + \varepsilon)^n$ for all but finitely many $n \in \mathbb{N}$.

The proof follows easily from the definition of the limit superior. By properties (i) and (ii), one can see that the exponential order provides suitable lower and upper bounds for a sequence. Note that the definition above is equivalent to $a_n = K^n\theta(n)$, where $\theta\colon \mathbb{N} \to \mathbb{R}$ is a subexponential factor, i.e. $\limsup |\theta(n)|^{1/n} = 1$. Examples of subexponential factors include $n^2, \sqrt{n}, \log(n)$. Combining this Corollary with the previous result gives following Proposition.

Proposition 3.4 (Naive bound)**.** *If $f\colon \mathbb{R} \to \mathbb{R}$ is analytic in the origin with non-negative Taylor coefficients $(a_n)_{n\in\mathbb{N}}$ and the closest non-removable singularity to the origin is the point $z = R$. Then, we have $a_n \bowtie \frac{1}{R^n}$ and there exists a subexponential factor θ with $a_n = \frac{1}{R^n}\theta(n)$.*

We can now determine the order of exponential growth of a sequence up to a subexponential factor simply by forming its generating function and determining the closest non-removable singularity to the origin. However, if a sequence is not of exponential order, the previous result does not provide a good estimate. So naturally we want to find a good approximation of the subexponential factor θ. Since A_n is dependent on the distance to the nearest singularity, it can be postulated that the subexponential factor depends on the type of singularity. This is indeed the case, as Flajolet and Sedgewick (see [2]) explore an expansion of meromorphic functions. The proof of the following theorem was only given as a rough sketch, but we will attempt to formalize the arguments.

Theorem 3.5 (Expansion of meromorphic functions)**.** *Let $U \subseteq \mathbb{C}$ be open and $f\colon U \to \mathbb{C}$ be meromorphic on U, containing the closed disc $\overline{D(0,R)} = \{z \in \mathbb{C}\colon |z| \leqslant R\}$, with poles at*

points $\alpha_1, \alpha_2, \ldots, \alpha_m$ *of order* $\operatorname{ord}_{\alpha_1}(f), \ldots, \operatorname{ord}_{\alpha_m}(f)$*. Further assume that* f *is holomorphic on* $\{z \in \mathbb{C} \colon |z| = R\}$ *and at* $z = 0$*. Then there exist* m *polynomials* $\{p_j\}_{j=1}^m$ *such that:*

$$a_n = \sum_{j=1}^{m} p_j(n)\alpha_j^{-n} + O\left(R^{-n}\right),$$

whereby the coefficients a_n *refer to the coefficients of the local Taylor expansion around the origin. Furthermore the identity* $\deg(p_j) = \operatorname{ord}_{\alpha_j}(f) - 1$ *holds true.*

Proof. Since f is holomorphic in the origin, there exists a neighborhood U containing $\overline{D(0,r)}$ for small enough $r > 0$ such that f is holomorphic on U. By Cauchy's integral formula (Notes 3, Corollary 20), we have

$$a_n = \frac{1}{2\pi i} \int_{\gamma_{0,r,\circlearrowleft}} \frac{f(z)}{z^{n+1}} \mathrm{d}z.$$

Furthermore, the function $\frac{f(z)}{z^{n+1}}$ only contains the origin as a singularity in the interior of the curve $\gamma_{0,r,\circlearrowleft}$. Applying the residue theorem (Notes 4, Theorem 22) yields

$$a_n = \frac{1}{2\pi i} \int_{\gamma_{0,r,\circlearrowleft}} \frac{f(z)}{z^{n+1}} \mathrm{d}z = \operatorname{Res}\left(\frac{f(z)}{z^{n+1}}; 0\right).$$

Now consider the integral $\frac{1}{2\pi i}\int_{\gamma_{0,R,\circlearrowleft}} \frac{f(z)}{z^{n+1}}\mathrm{d}z$ and the curve $\gamma_{0,R,\circlearrowleft}$. Once again applying the residue theorem gives

$$\begin{aligned} \frac{1}{2\pi i} \int_{\gamma_{0,R,\circlearrowleft}} \frac{f(z)}{z^{n+1}} \mathrm{d}z &= \operatorname{Res}\left(\frac{f(z)}{z^{n+1}}; 0\right) + \sum_{j=1}^{m} \operatorname{Res}\left(\frac{f(z)}{z^{n+1}}; \alpha_j\right) && (1) \\ &= a_n + \sum_{j=1}^{m} \operatorname{Res}\left(\frac{f(z)}{z^{n+1}}; \alpha_j\right). && (2) \end{aligned}$$

Consider the terms $\operatorname{Res}\left(\frac{f(z)}{z^{n+1}}; \alpha_j\right)$ for $j = 1, \ldots, m$. By Notes 4, Exercise 11 (ii), we have $\operatorname{ord}_{\alpha_j}(\frac{f(z)}{z^{n+1}}) = \operatorname{ord}_{\alpha_j}(f) + \operatorname{ord}_{\alpha_j}(\frac{1}{z^{n+1}}) = \operatorname{ord}_{\alpha_j}(f)$. Applying Theorem 1.4 (Chapter 3, Stein-Shakarchi [8])

$$\operatorname{Res}\left(\frac{f(z)}{z^{n+1}}; \alpha_j\right) = \lim_{z \to \alpha_j} \frac{1}{(\operatorname{ord}_{\alpha_j}(f) - 1)!} \left(\frac{d}{dz}\right)^{\operatorname{ord}_{\alpha_j}(f)-1} (z - \alpha_j)^{\operatorname{ord}_{\alpha_j}(f)} \frac{f(z)}{z^{n+1}}.$$

If we define $q_j(n) := \operatorname{Res}\left(\frac{f(z)}{z^{n+1}}; \alpha_j\right)$, then one can write $q_j = A_j \left(\alpha_j^{-1} n\right)^{\operatorname{ord}_{\alpha_j}(f)-1} \alpha_j^{-n} + O\left(n^{\operatorname{ord}_{\alpha_j}(f)-2}\right)$ for some constant A_j, which can be seen as follows. First consider Theorem 1.3 (Chapter 3, Stein-Shakarchi [8]), which states that for a function g with a pole z_0 of order k, there exists a holomorphic function H such that

$$g(z) = \frac{a_{-k}}{(z - z_0)^k} + \frac{a_{-k+1}}{(z - z_0)^{k-1}} + \cdots + \frac{a_{-1}}{(z - z_0)} + H(z).$$

Applying this theorem to f yields

$$q_j(n) = \lim_{z\to\alpha_j} \frac{1}{(\operatorname{ord}_{\alpha_j}(f)-1)!}\left(\frac{d}{dz}\right)^{\operatorname{ord}_{\alpha_j}(f)-1} \frac{a_{-\operatorname{ord}_{\alpha_j}(f)}}{z^{n+1}} + \frac{a_{-\operatorname{ord}_{\alpha_j}(f)+1}(z-\alpha_j)}{z^{n+1}} + \cdots + \frac{H(z)(z-\alpha_j)^{\operatorname{ord}_{\alpha_j}}}{z^{n+1}}$$
$$\approx \lim_{z\to\alpha_j} A_j \frac{n^{\operatorname{ord}_{\alpha_j}(f)-1}}{z^{n+1+\operatorname{ord}_{\alpha_j}(f)}} + O\left(n^{\operatorname{ord}_{\alpha_j}(f)-2}\right) = A_j(\alpha_j^{-1}n)^{\operatorname{ord}_{\alpha_j}(f)}\alpha_j^{-n} + O\left(n^{\operatorname{ord}_{\alpha_j}(f)-2}\right),$$

whenever $n \gg \operatorname{ord}_{\alpha_j}(f)$ and the derivative of the first term dominates the other terms, which is the case, if $|\alpha_j| \leqslant 1$. Defining $p_j(n) := q_j(n)\alpha_j^n$ provides the wanted polynomials. Lastly, bounding the left hand side of (1) and rearranging (2) yields

$$\left|\frac{1}{2\pi i}\int_{\gamma_{0,R,\circlearrowleft}} \frac{f(z)}{z^{n+1}}\mathrm{d}z\right| \leqslant \left|\frac{2\pi R}{2\pi i}\right| \sup_{|z|=R}\left|\frac{f(z)}{z^{n+1}}\right| = \frac{\sup_{|z|=R}|f(z)|}{R^n} = O(R^{-n}), \tag{3}$$

whereby the supremum is attained and finite, since f is holomorphic and therefore continuous on the compact set $\overline{D(0,R)}$. □

In the proof we used the fact that the singularities satisfy $|\alpha_j| \leqslant 1$ for $j = 1, \ldots, m$, which was not mentioned by Flajolet and Sedgewick in (pg.258 Theorem IV.10, [2]). This is however not a strong additional assumption if we want to find asymptotic bounds, as Proposition 3.4 justifies that only these singularities have some asymptotic relevancy, though dominated by the factor α_1^{-n}, whereby α_1 is the smallest singularity.

The upper bound (3) also motivates an alternative method is to approximate the coefficients $a_{n\in\mathbb{N}}$ directly using the Cauchy-integral theorem, as it provides an explicit representation of a_n. This method however requires stronger assumptions.

Theorem 3.6 (Saddle point bounds)**.** *Let $U \subseteq \mathbb{C}$ be open and $f\colon U \to \mathbb{C}$ be analytic in the disc $D(0,R)$ with $R > 0$. Then we have*

$$a_n \leqslant \inf_{r\in(0,R)} \frac{\sup_{|z|=r}|f(z)|}{r^n}. \tag{4}$$

Proof. This is a simple application of Cauchy's integral formula (Notes 3, Corollary 20) and bounding the contour integral identically to (3)

$$a_n = \left|\frac{1}{2\pi i}\int_{\gamma_{0,r,\circlearrowleft}} \frac{f(z)}{z^{n+1}}dz\right| \leqslant \left|\frac{2\pi r}{2\pi i}\right| \sup_{|z|=r}\left|\frac{f(z)}{z^{n+1}}\right| = \frac{\sup_{|z|=r}|f(z)|}{r^n},$$

whereby we used the fact that a_n is real and non-negative. Additionally, Cauchy's integral formula holds for any $0 < r < R$, as the curves are homotopic. The statement follows from taking the infimum on the right hand side of the inequality. □

Note that for functions with real and non-negative Taylor-coefficients, we can rewrite the right hand side of (4) as

$$a_n \leqslant \inf_{r\in(0,R)} \frac{|f(r)|}{r^n}.$$

4 Application

We will now return to the examples that we started with and apply the discussed methods.

Example 4.1 (Fibonacci sequence)**.** As established in Example 2.2, the generating function for the Fibonacci sequence $(F_n)_{n\in\mathbb{N}}$ is given by $G(z) = \frac{z}{1-z-z^2}$, which has simple poles in the points $\alpha_1 := \frac{-1+\sqrt{5}}{2}$ and $\alpha_2 := \frac{-1-\sqrt{5}}{2}$ both of order 1. Additionally $|\alpha_1| < |\alpha_2|$ holds.

Naive bound: By Proposition 3.4, the sequence is of exponential order $\frac{1}{\alpha_1^n}$, i.e.

$$F_n \bowtie \left(\frac{2}{-1+\sqrt{5}}\right)^n.$$

Expansion method: Using Theorem 3.5, we have

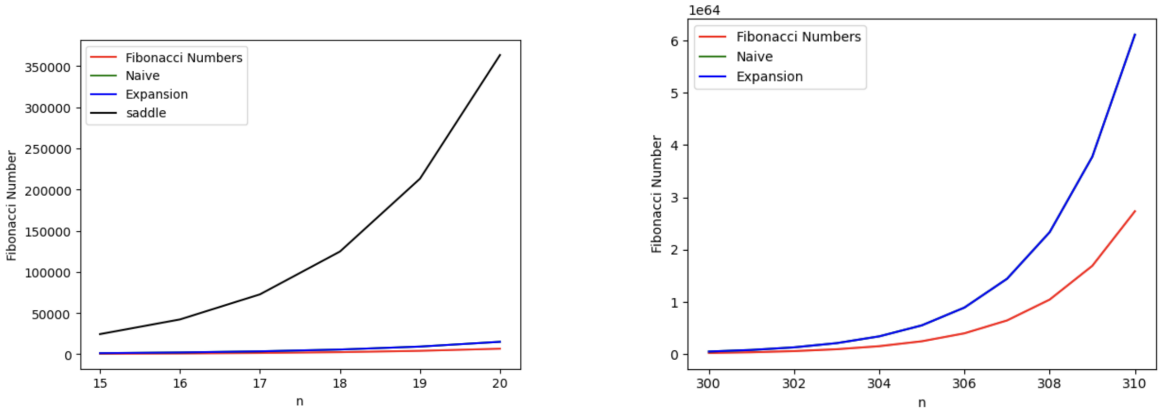
$$F_n \approx \frac{p_1(n)}{\alpha_1^n} - \frac{p_2(n)}{\alpha_2^n},$$

whereby p_1 and p_2 are constant polynomials, since we have simple poles. Asymptotically the approximation is the same as the naive bound, as the second pole α_2 has no influence on the asymptotic behavior, as $\left|\frac{1}{\alpha_2}\right| < 1$. From this example, we can clearly see that the relevant poles are those with modulus less than 1.

Saddle-point method: Using the saddle-point method, we only consider $R < \alpha_1$, in order to satisfy the assumptions of Proposition 3.6.

$$F_n \leqslant \inf_{r\in(0,\alpha_1)} \frac{\sup_{|z|=r}|f(z)|}{r^n} \leqslant \inf_{r\in(0,\alpha_1)} \left|\frac{r}{r^n(1-r-r^2)}\right|.$$

The infimum on the right hand side can be compute easily by considering its derivative. For the sake of this report, these calculations will be left out and Figure 1 show the approximations graphically.



(a) 15-20th Fibonacci Number

(b) 300-310th Fibonacci Number

Figure 1: Approximation of Fibonacci Numbers

For this example the saddle-point method has not provided very accurate bounds. One can see from the bound that this method is most effective, if f does not contain any singularities or singularties that are very far from the origin such that $R \gg 0$. From the above discussion, the Naive and the Expansion method also coincide asymptotically. Nevertheless, if there are multiple singularities less than 1, the Expansion method should provide a better estimate. For $n = 200$, the Naive and Expansion method incur a relative error of approximately 1.24, whereas the Saddle-point method has a relative error of around 542.

Example 4.2 (Recursion equation)**.** Returning to the sequence $(a_n)_{n\in\mathbb{N}}$ from Example 2.4 with generating function

$$G(z) = \frac{1 - 10z + 44z^2 - 84z^3 + 80z^4 - 32z^5}{(1-z)(1-2z)^5}.$$

We have two poles, namely $\alpha_1 = \frac{1}{2}$ of order 1 and $\alpha_2 = \frac{1}{2}$ of order 5.

Naive method: Similarly, we obtain the exponential order $\frac{1}{\alpha_1^n}$, so

$$a_n \bowtie 2^n$$

Expansion method: By Theorem 3.5, we have

$$a_n \approx \frac{p_1(n)}{\alpha_1^n} + \frac{p_2(n)}{\alpha_2^n} = 2^n p_1(n) + p_2(n)$$

for a polynomial p_1 of order 4 and constant polynomial p_2. For simplicity, we will assume that $p_1 = n^4$ and $p_2 = 1$, but more exact approximations can be made using the formula in the proof of Theorem 3.5.

Saddle-point method: Lastly using Theorem 3.6

$$a_n \leqslant \inf_{r\in(0,\alpha_1)} \frac{\sup_{|z|=r} |G(z)|}{r^n} \leqslant \inf_{r\in(0,\alpha_1)} \left| \frac{1 - 10r + 44r^2 - 84r^3 + 80r^4 - 32r^5}{r^n(1-r)(1-2r)^5} \right|.$$

For $n = 100$, the relative error for the Saddle-point method is around 242, whereas the relative error for the naive method is around 1 and for the expansion method 0.98. This supports our hypothesis that the expansion method does indeed offer a slightly higher degree of accuracy. Figure 2 shows the approximation using the expansion method.

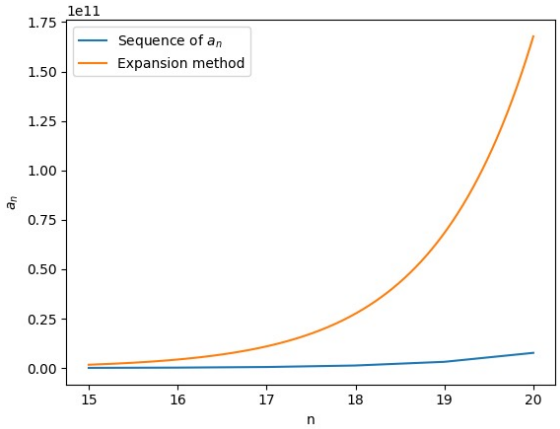


Figure 2: Approximation using Expansion method

5 Further remarks

Nevertheless the described methods in this report are not always effective, especially if the singularities of the generating functions lie on the unit circle and cannot be analyzed easily. The following example shows this.

Example 5.1 (Integer Partitions)**.** The number of partitions $p(n)$ of an integer n is an unordered combination of positive integers that sum to n. The generating function can be expressed as (see [5] for detailed derivation)

$$G(z) = \sum_{n=0}^{\infty} p(n) z^n = \prod_{k=0}^{\infty} \frac{1}{1 - z^k}.$$

The generating function has a singularity in 1 and so the naive bound provides the exponential order of 1, which is of little meaning. From the expression above it is difficult to determine the order of the pole or even if it is an essential singularity. For this singularity analysis is useful, which is another topic addressed in "Analytic Combinatorics" [2]. Nevertheless, this problem can still be tackled using contour integration and other methods. An interested reader is encouraged to see Stein-Shakarchi [8], Appendix A, Chapter 4) for a detailed derivation of the asymptotic bound.

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